

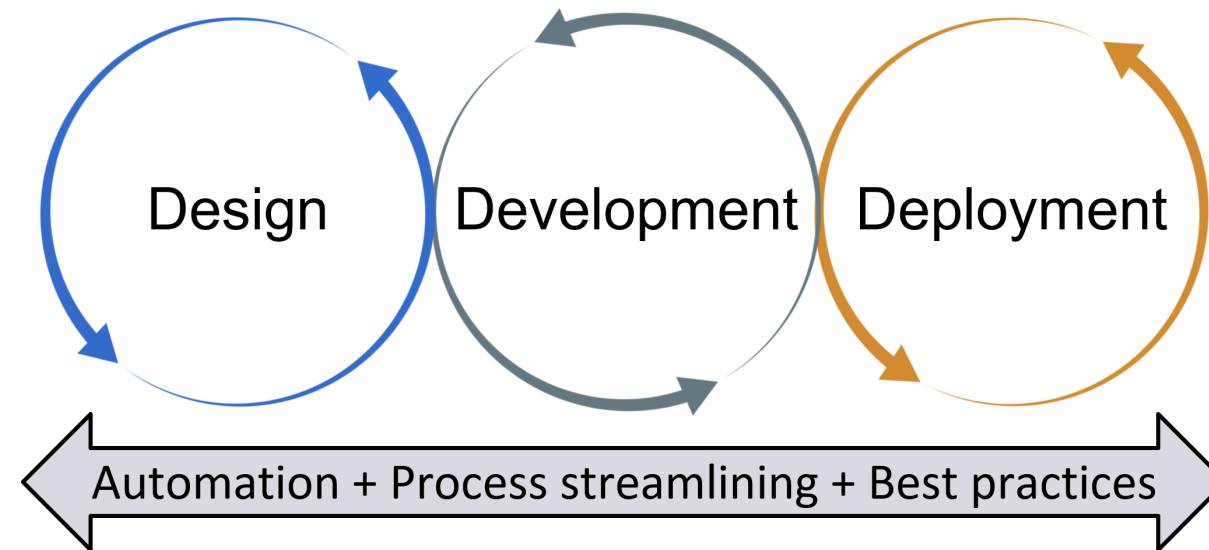
Introducing full automation and best practices to MLOps

FULLY AUTOMATED MLOPS



Arturo Opsetmoen Amador
Senior Consultant - Machine Learning

The MLOps lifecycle

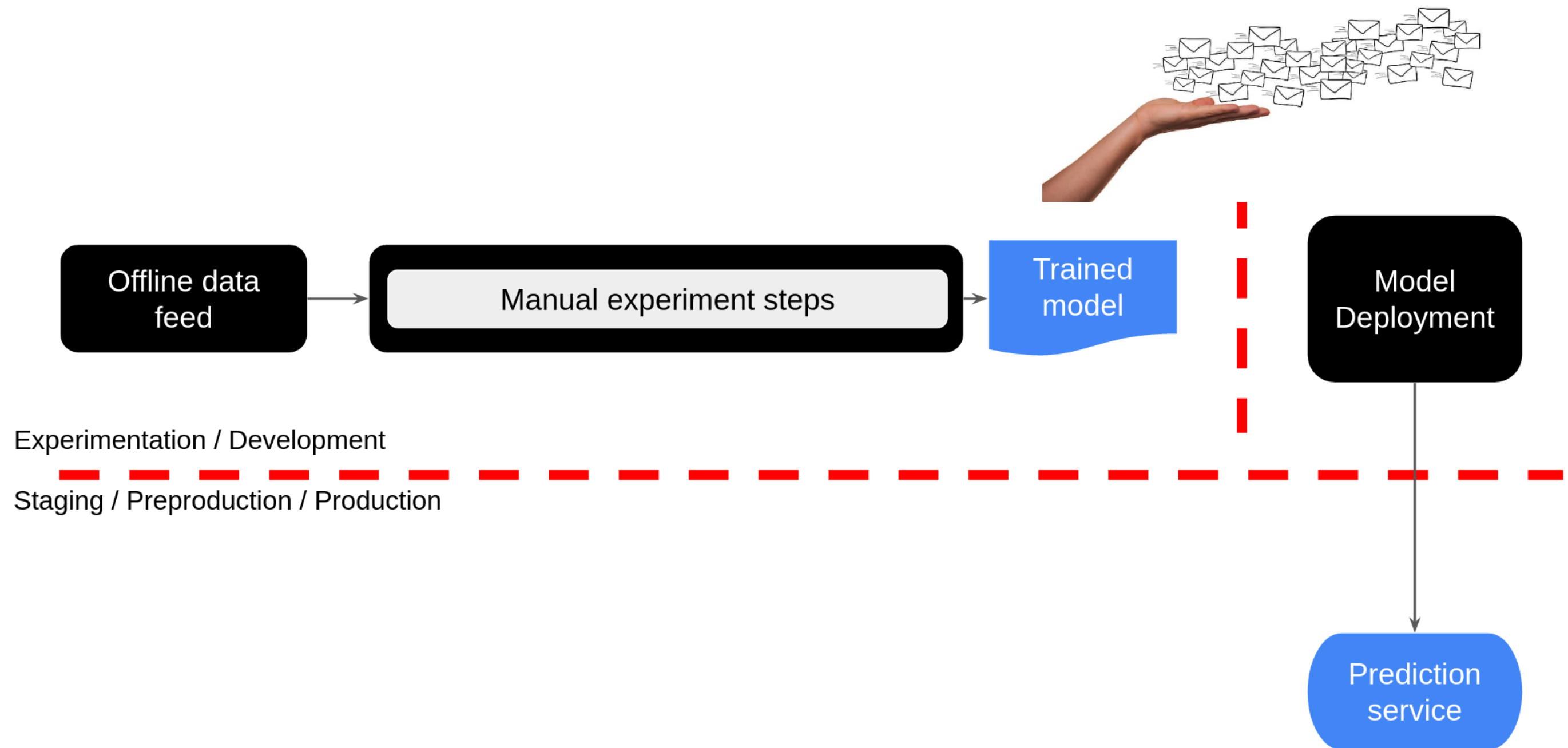


Maturity levels in MLOps

The maturity levels include:

- Manual ML workflow
- Semi-automated ML workflow
- Fully automated ML workflow

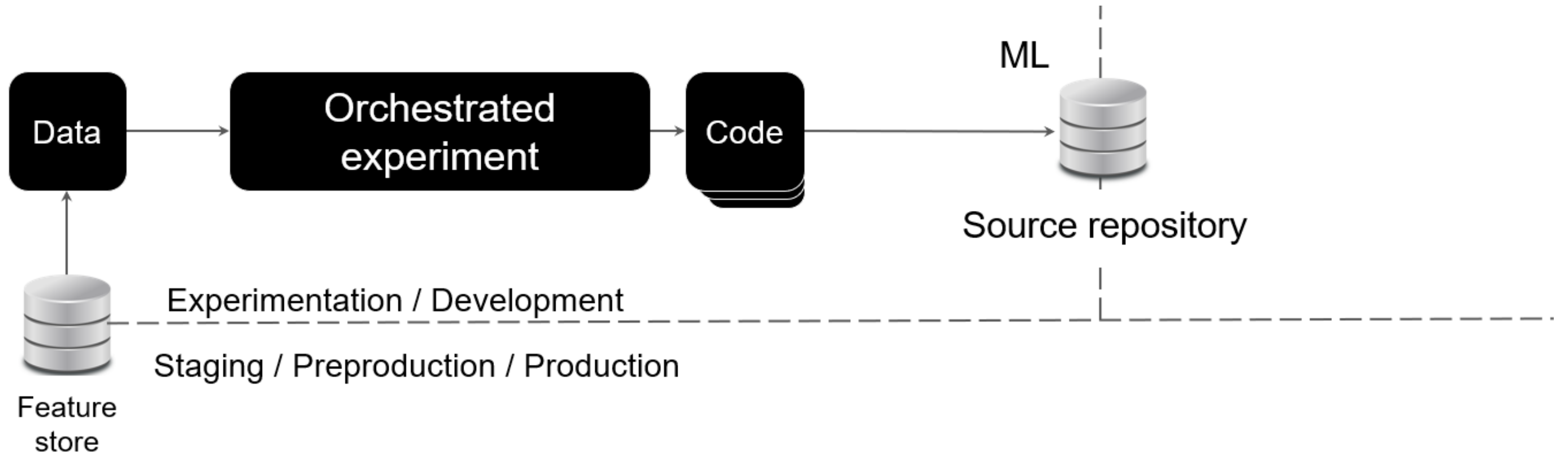
Manual ML workflow - Ad hoc experimentation



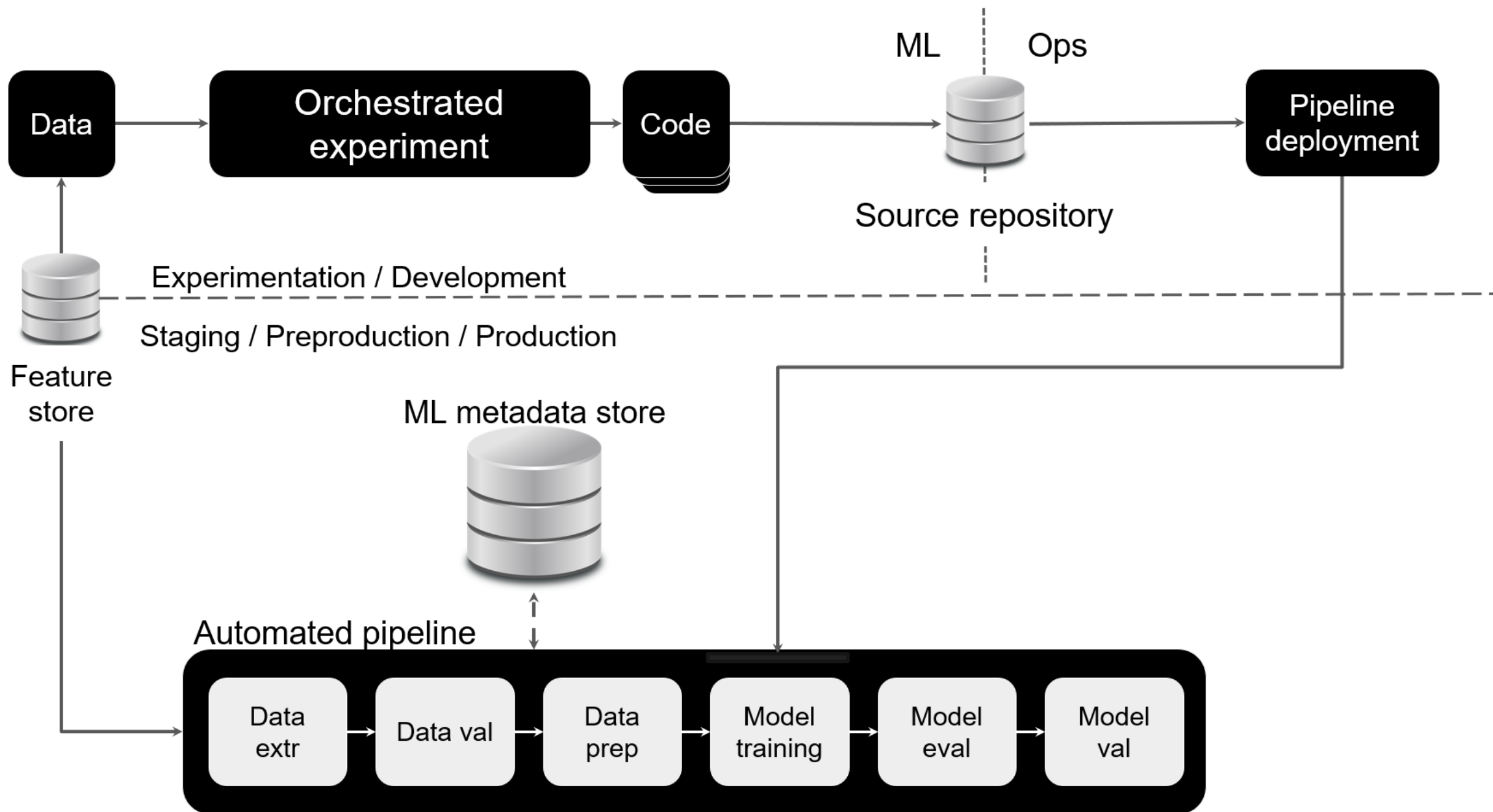
Semi-automated ML workflow



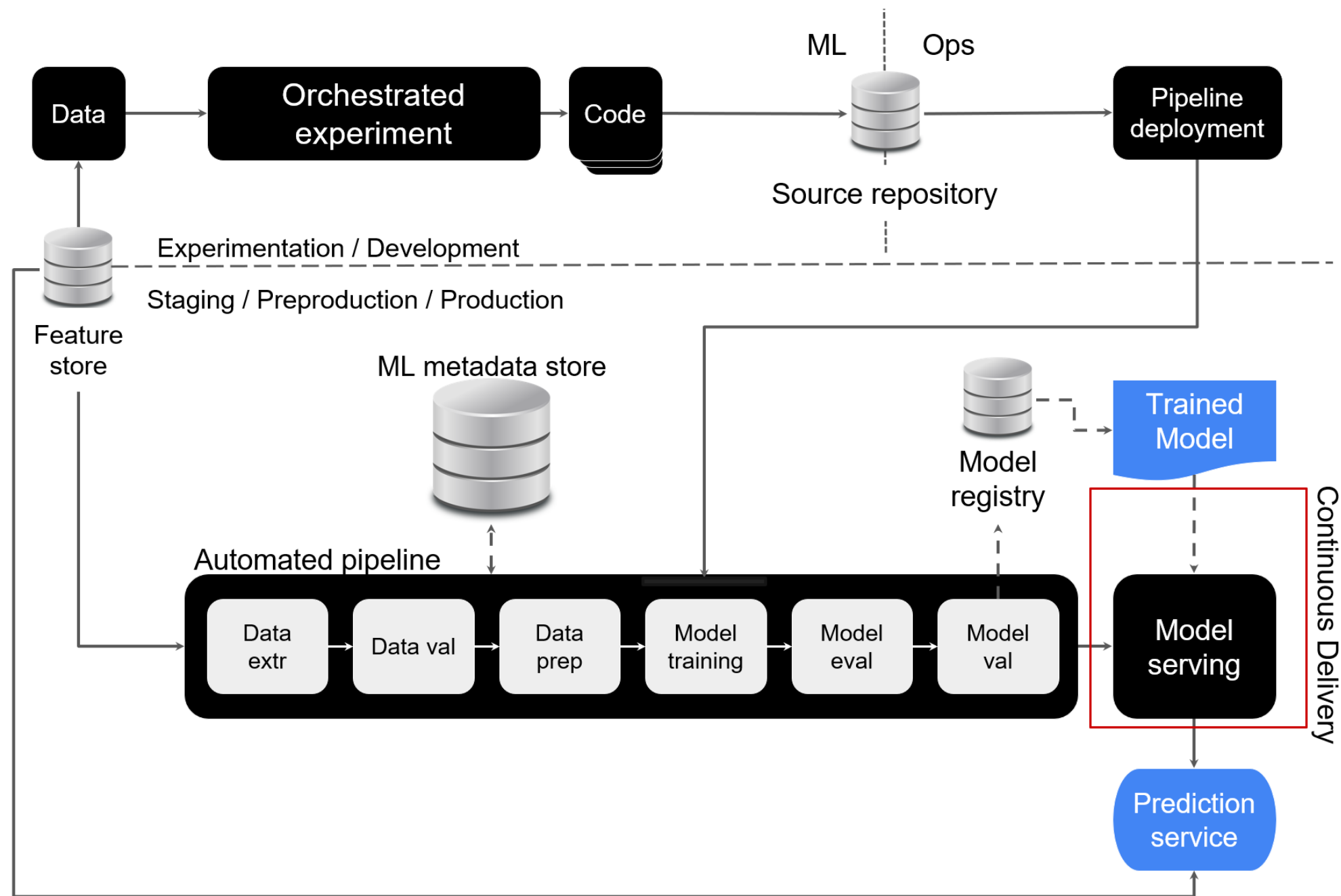
Semi-automated ML workflow



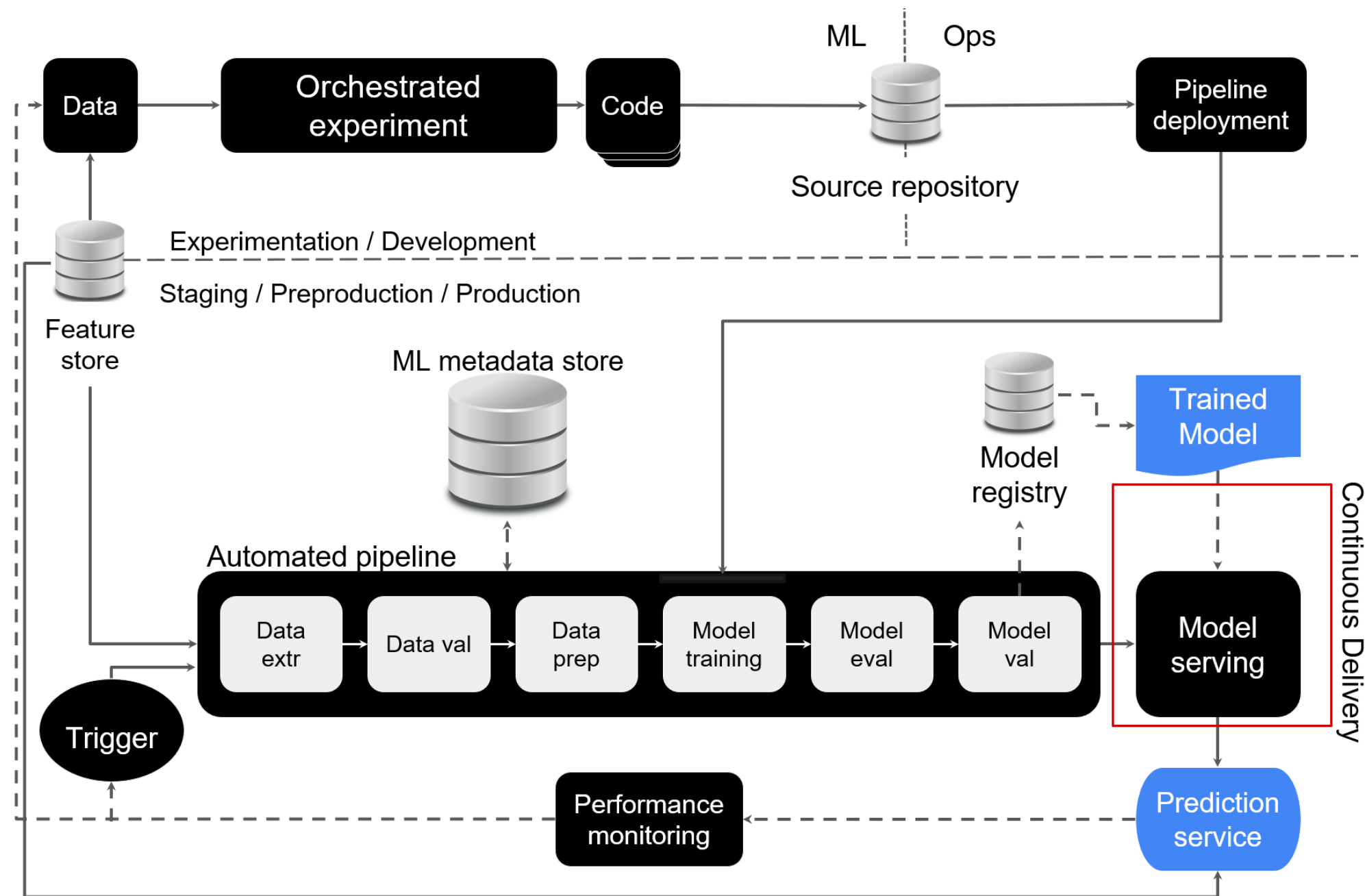
Semi-automated ML workflow



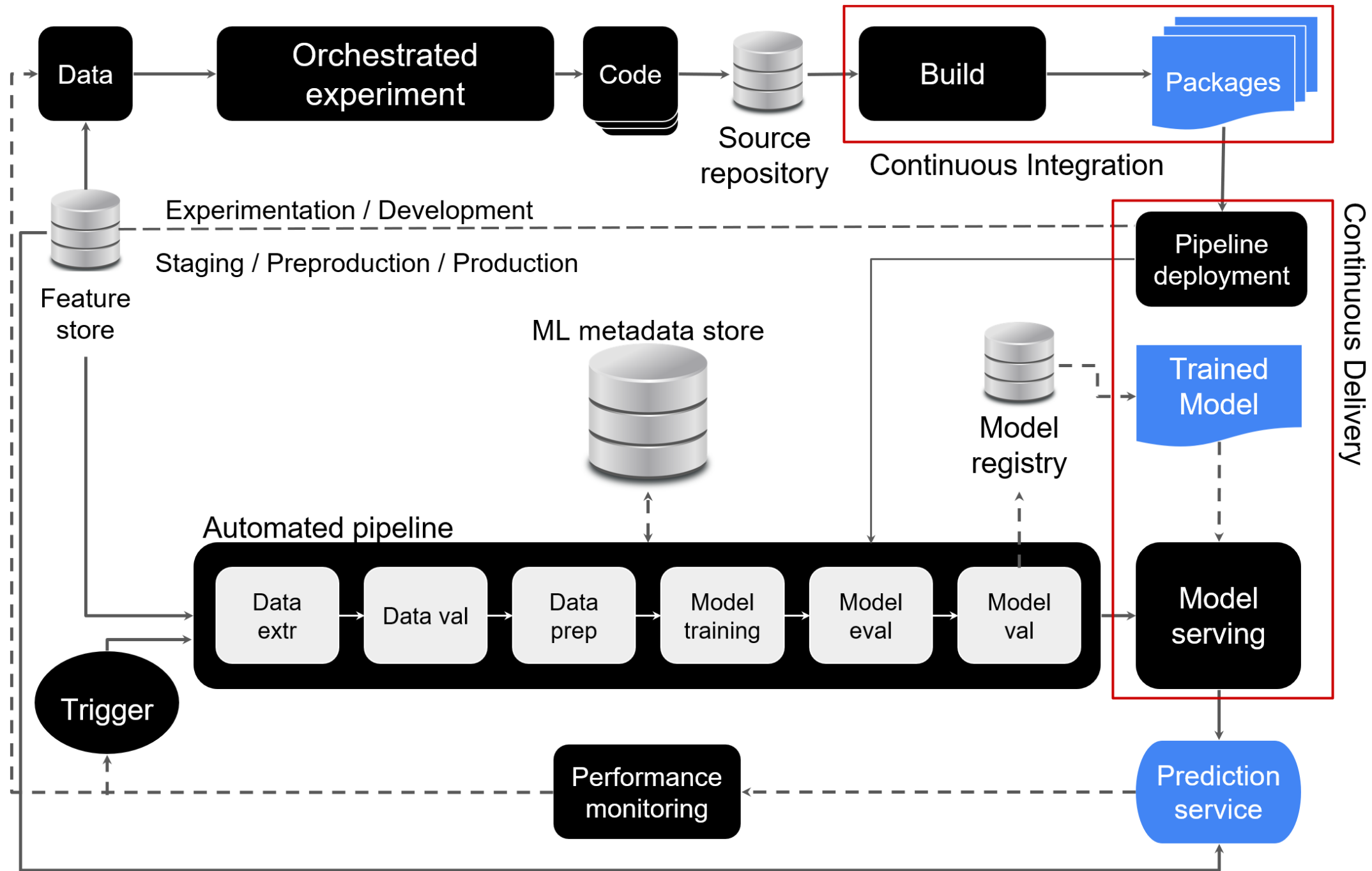
Semi-automated ML workflow



Semi-automated ML workflow



Fully automated ML workflow



Automation in the ML life cycle - Design

	Activity in Workflow	Fully Automated
Design & Planning	Business understanding	⊖
	Data understanding	⊖
	System design	⊖

Good practices:

- Apply reproducible processes
- Write detailed documentation

Automation in the ML life cycle - Development

	Activity in Workflow	Fully Automated
Development & Experimentation	Developing PoCs	⊖
	Data engineering	⊖
	Model development	⊖

Good practices:

- Remember we are developing software
- Using version control
- Use orchestration tools

Automation in the ML life cycle - Operations

	Activity in Workflow	Fully Automated
Deployment & Operations	Testing	✓
	Versioning	✓
	Monitoring	✓

Use:

- Automated testing
- CI/CD/CT/CM

Let's practice!
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The automation, monitoring, incident response pattern

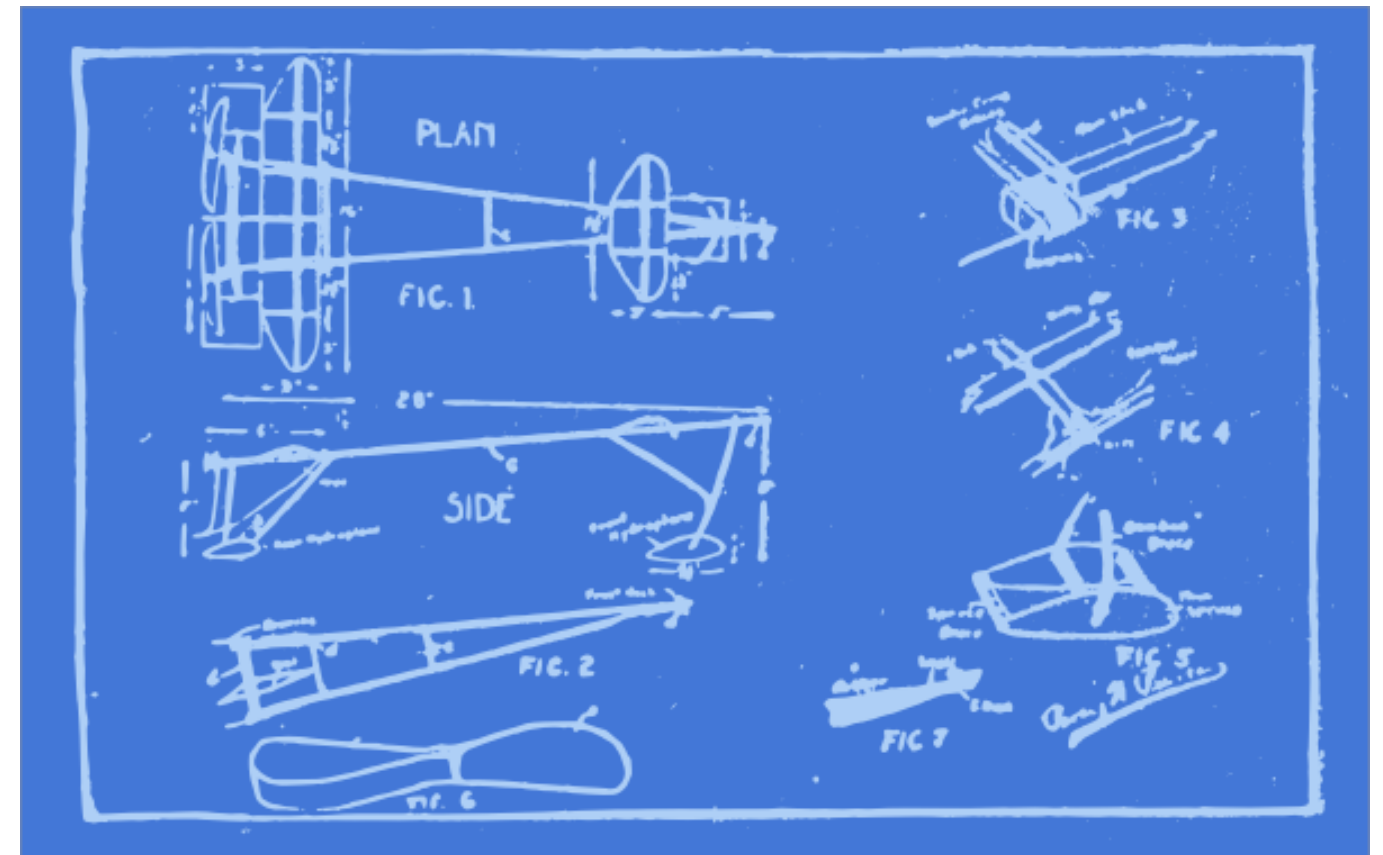
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What is a software design pattern?

A general, reusable solution to a commonly occurring problem...



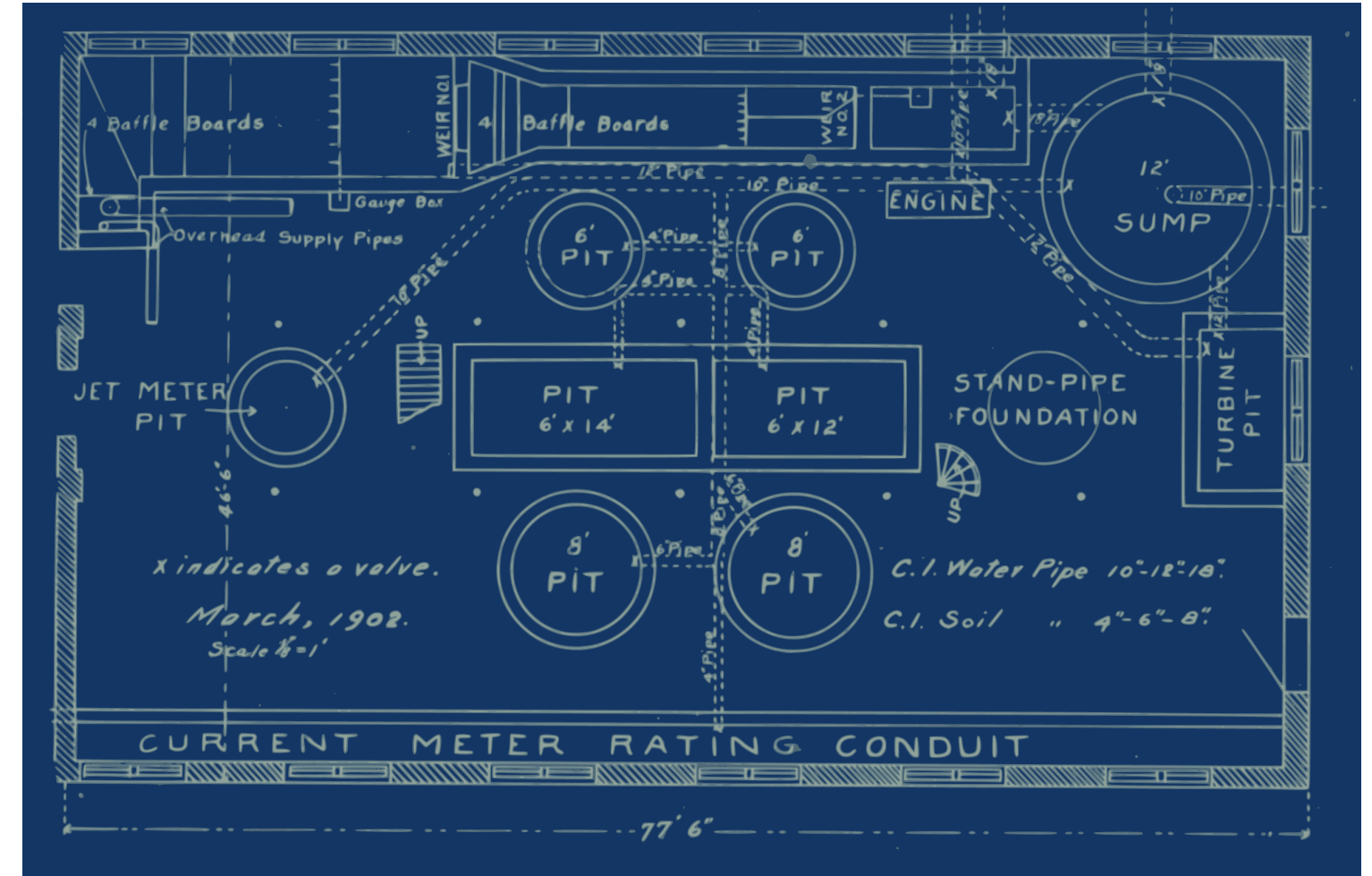
Automate, monitor, respond



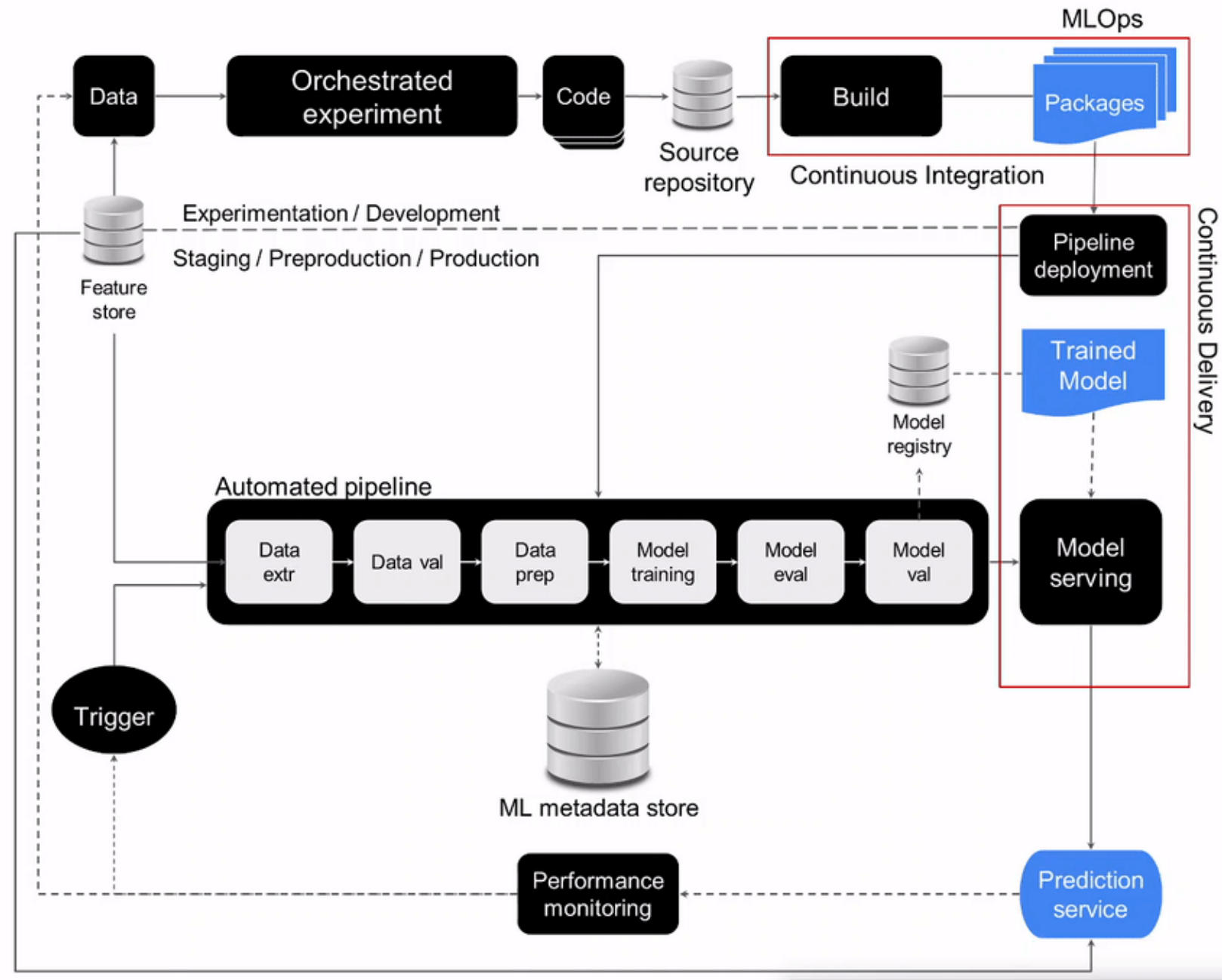
- Improves the reliability of the ML systems we design

Three examples of a design pattern in MLOps

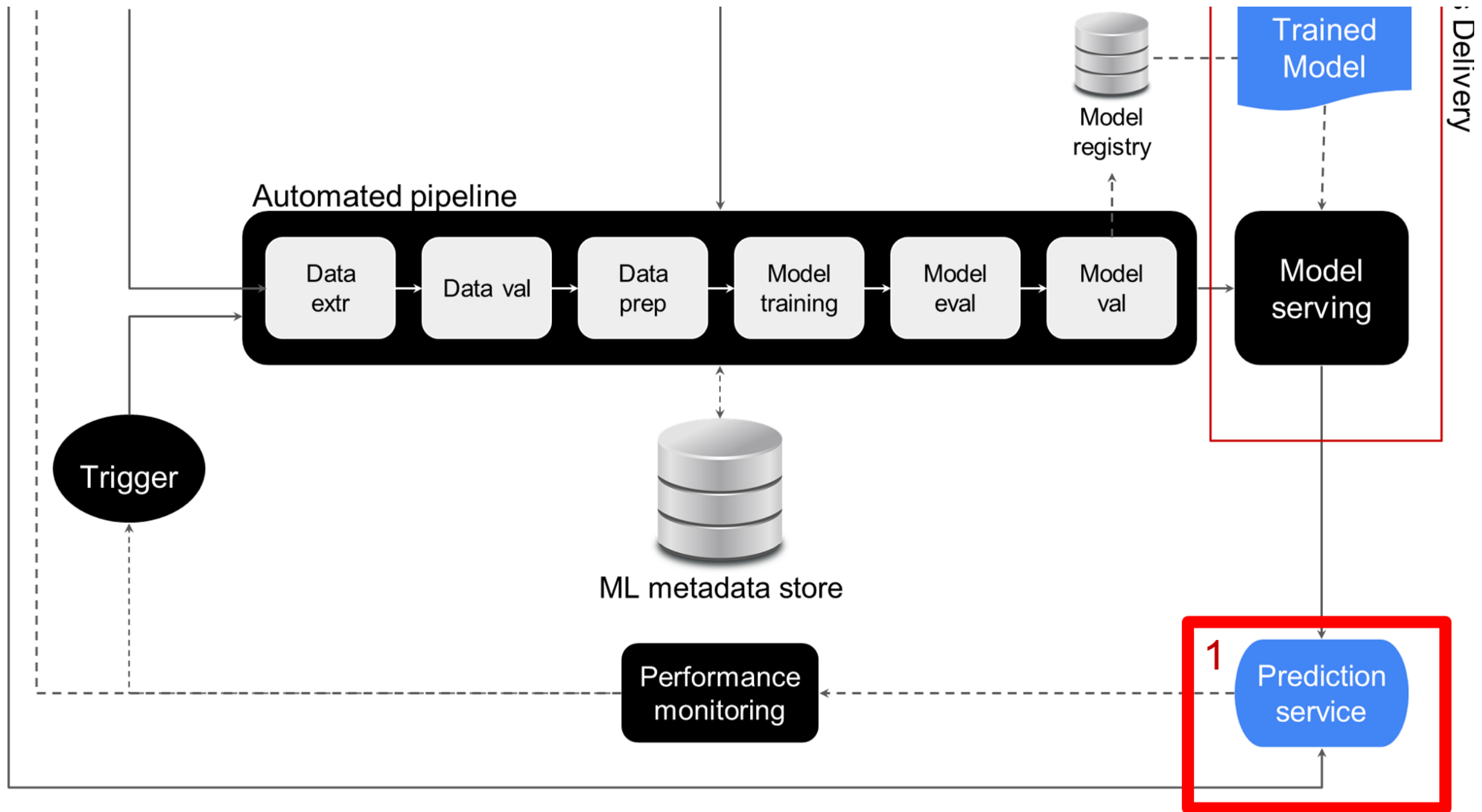
1. Automated Model Retraining
2. Model Rollback
3. Feature Imputation



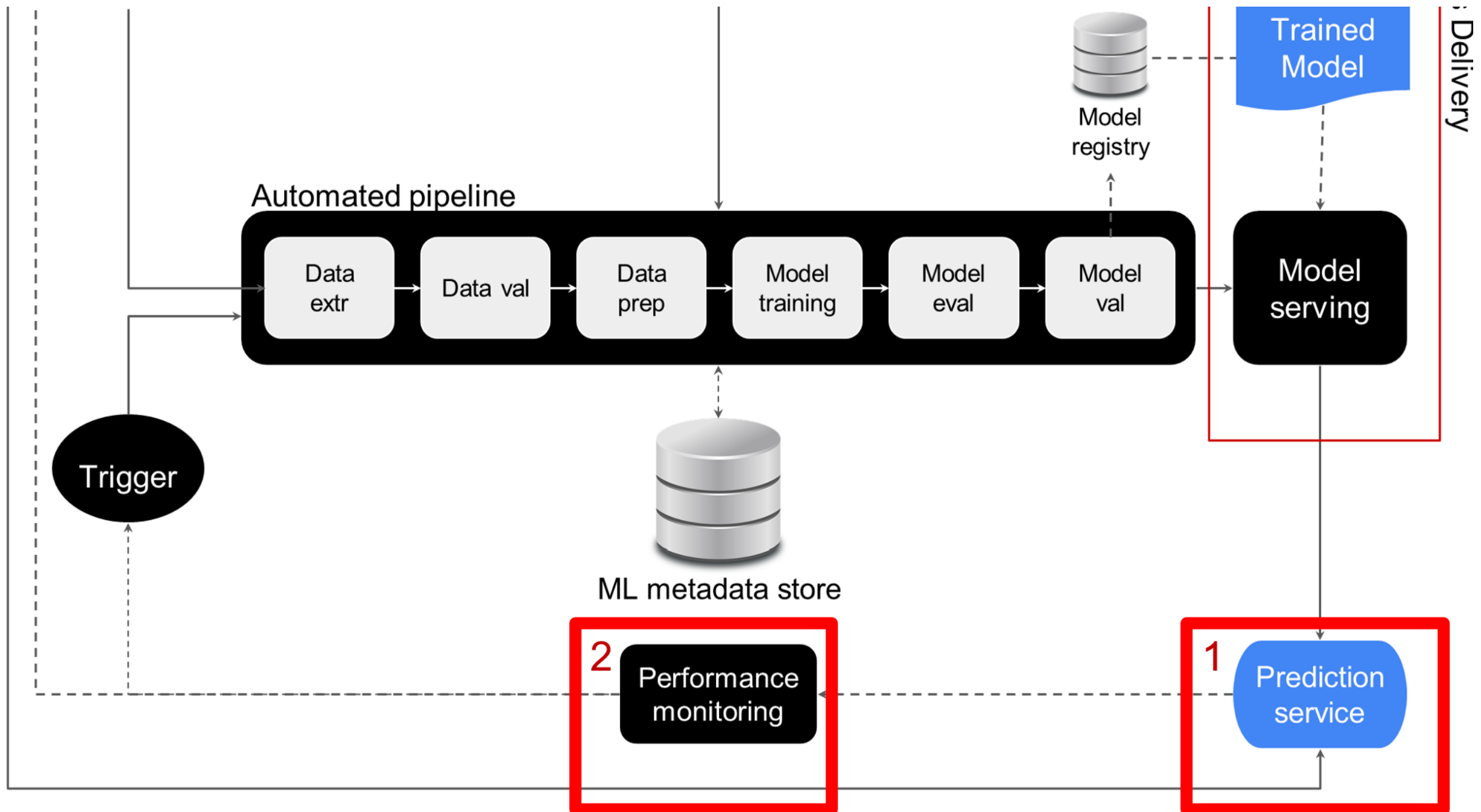
1. Automated model retraining



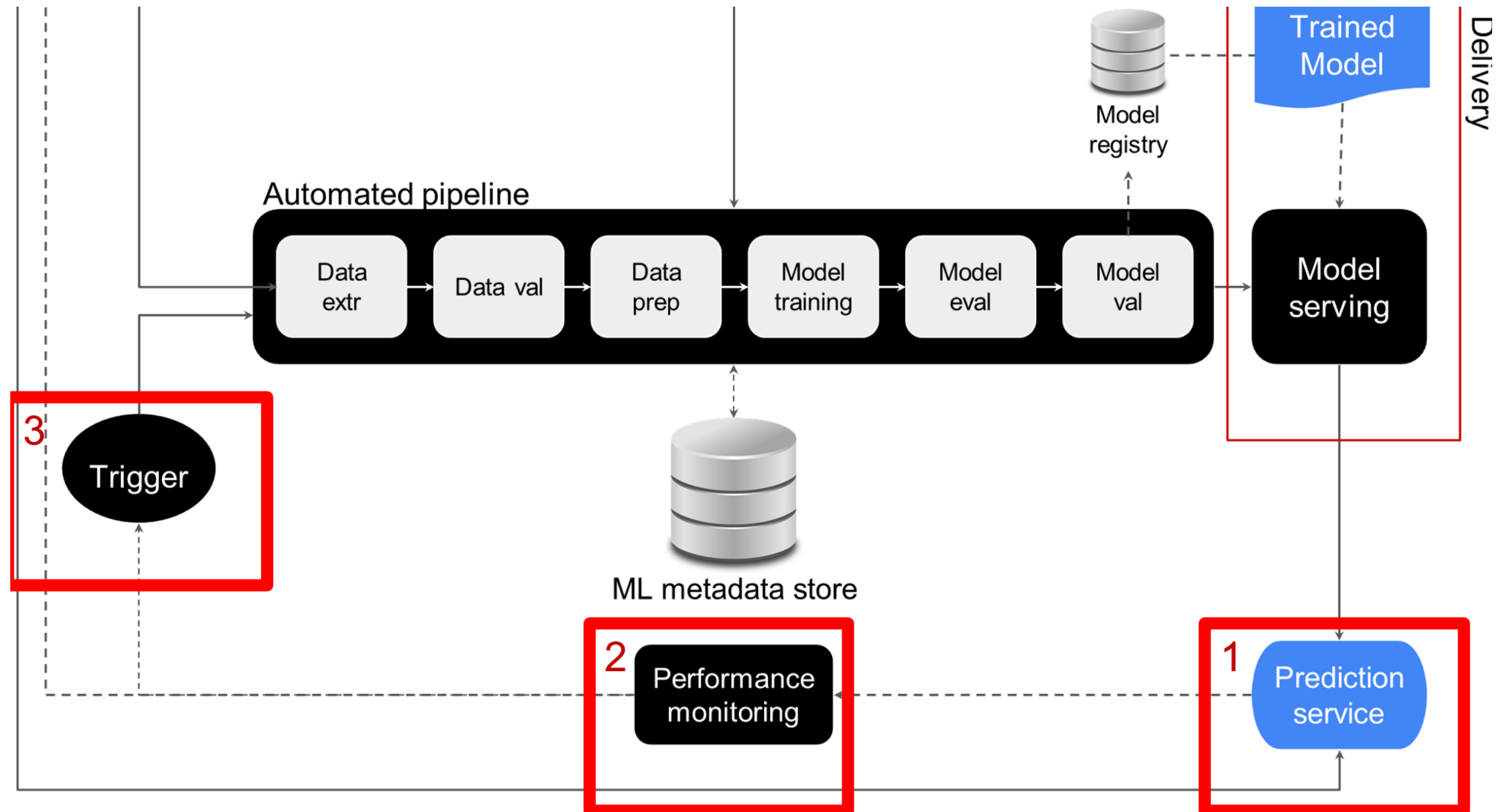
1. Automated model retraining - running predictions



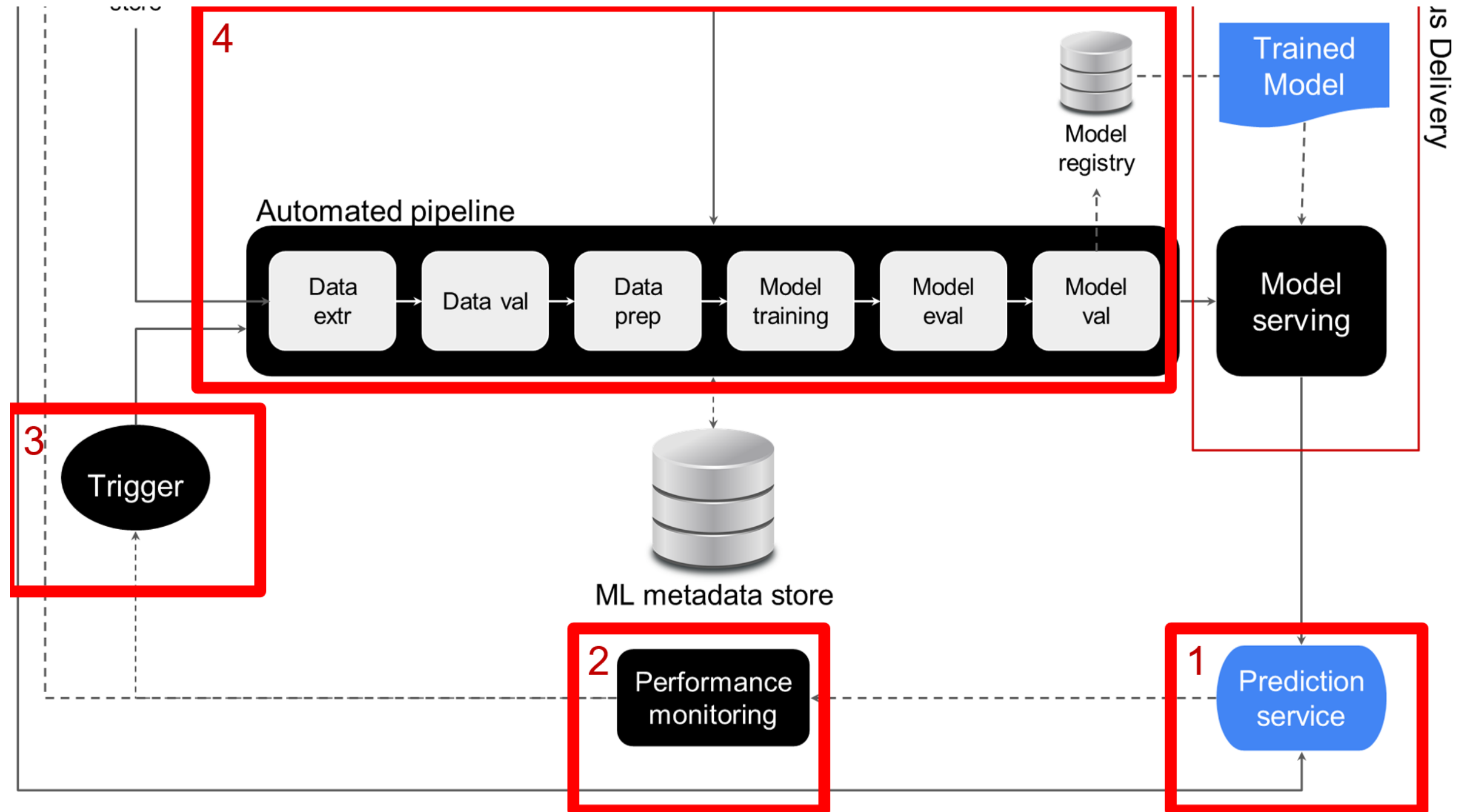
1. Automated model retraining - Monitoring



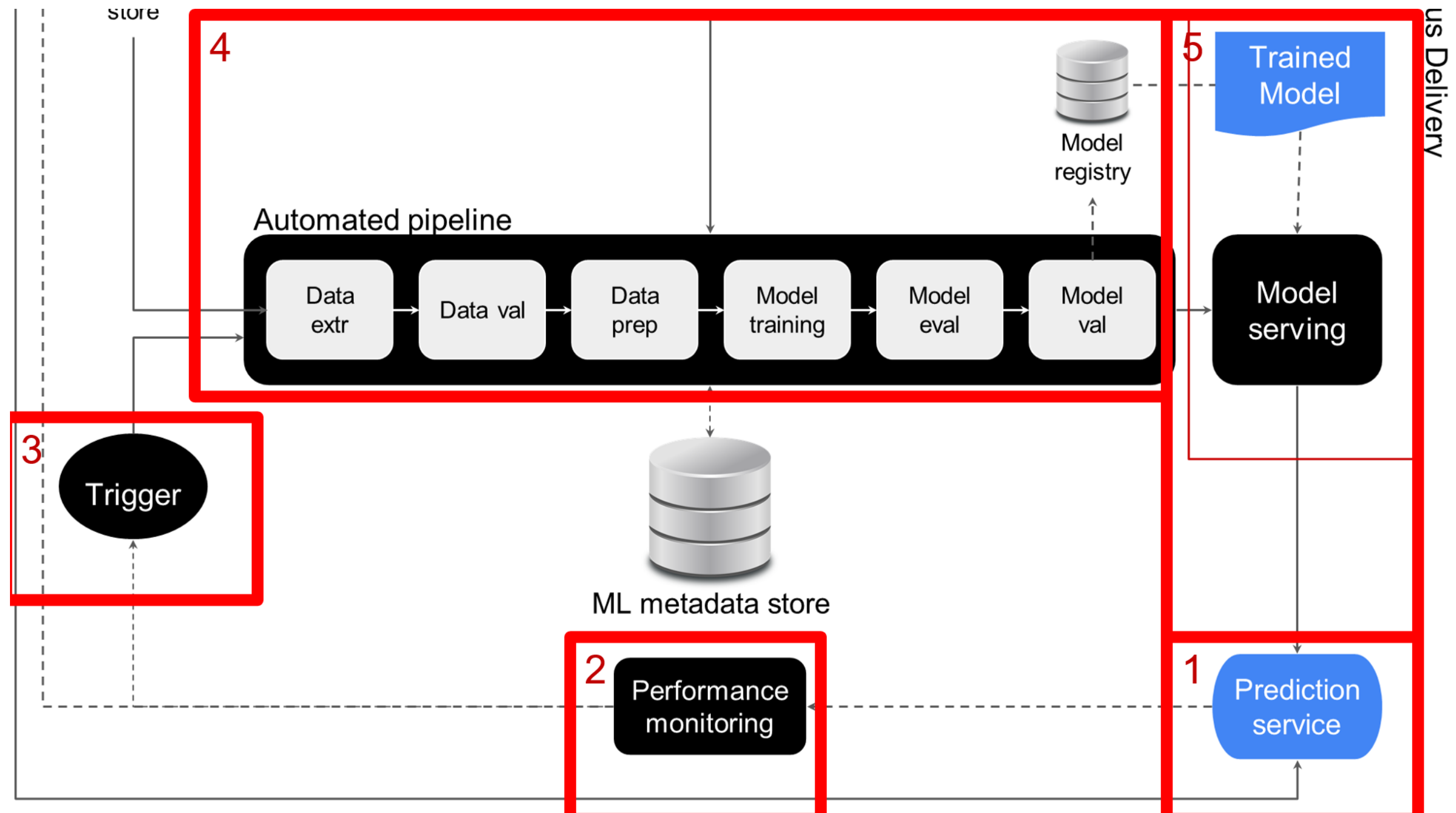
1. Automated model retraining - Trigger



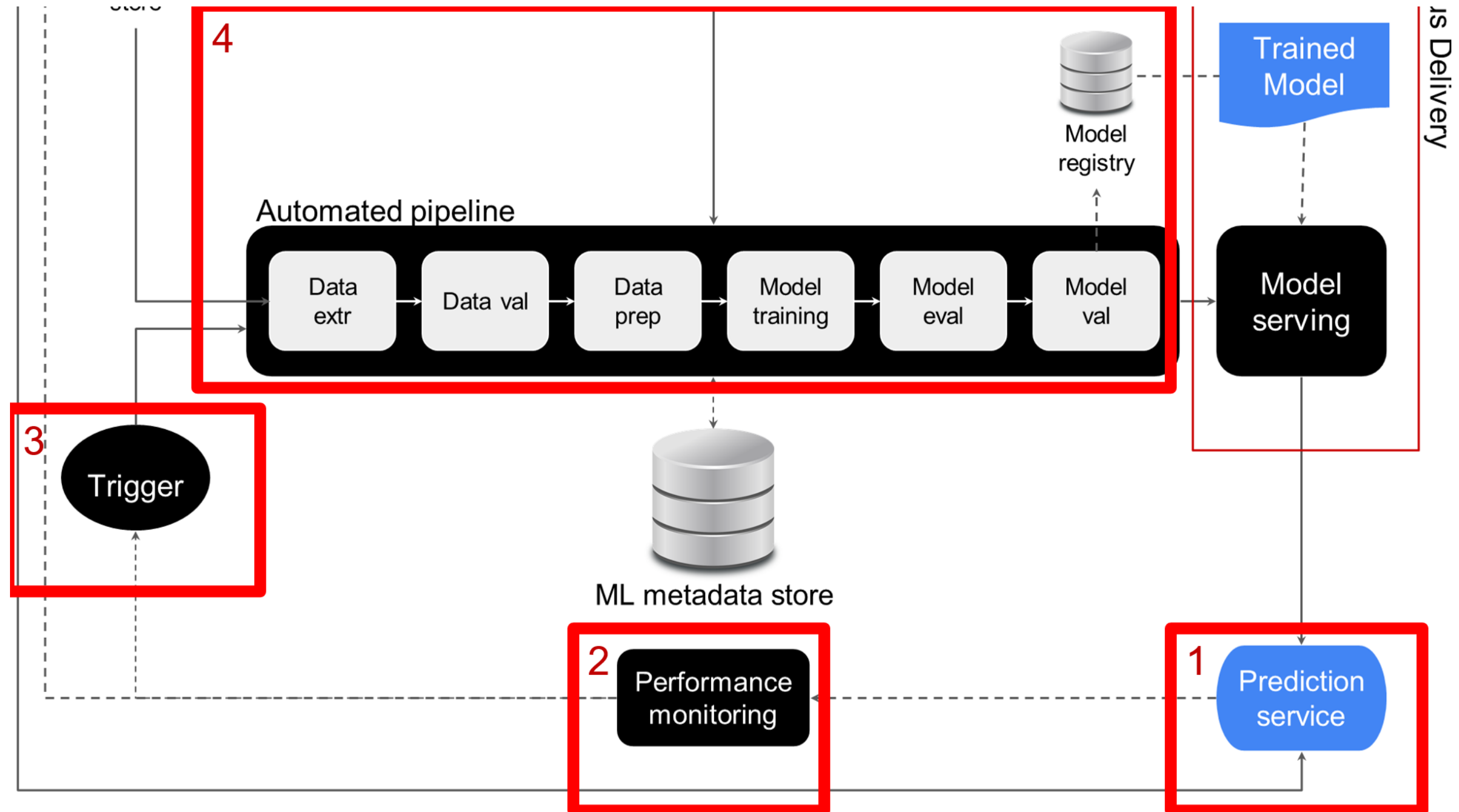
1. Automated model retraining - Automated pipeline



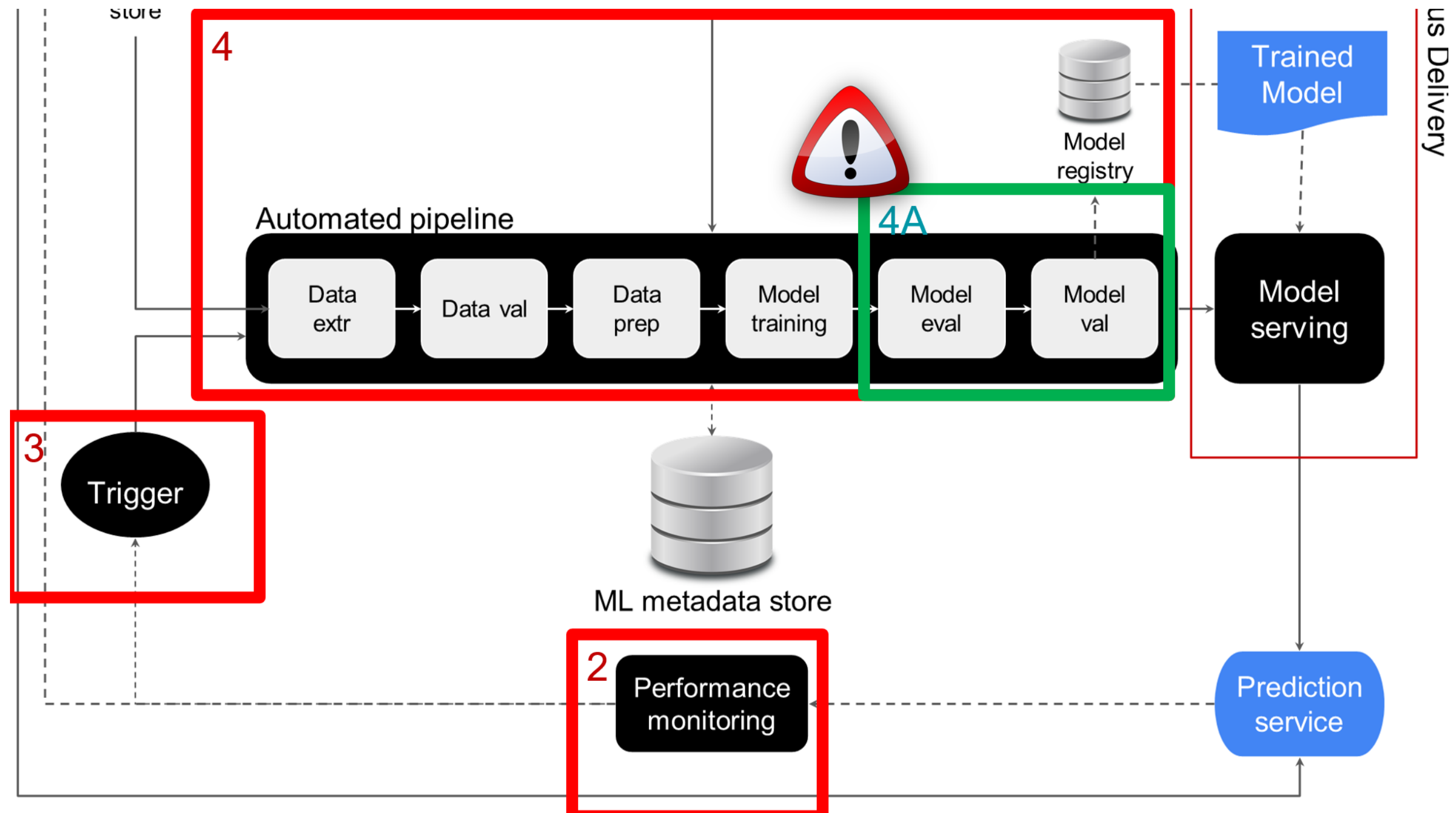
1. Automated model retraining - Deployment



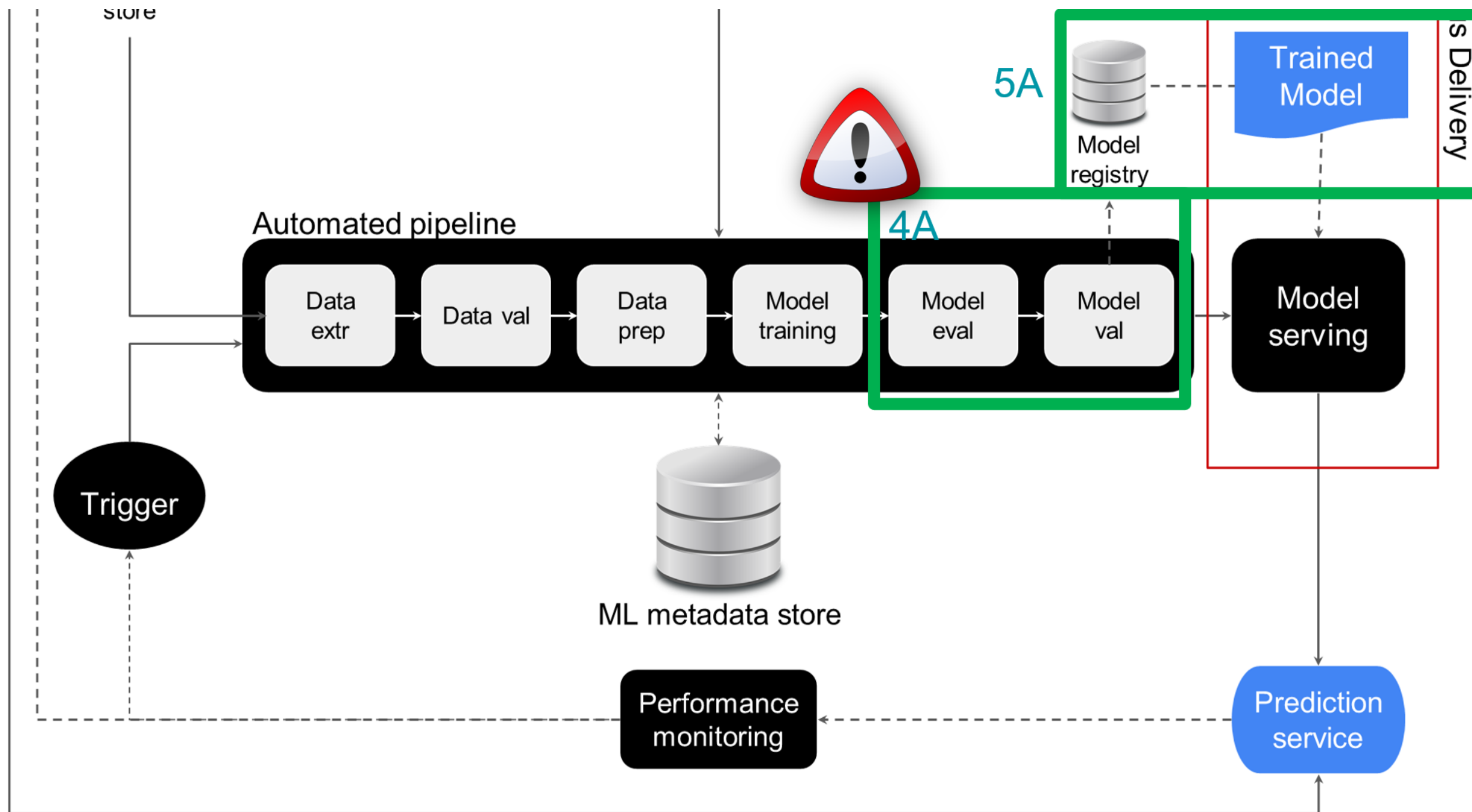
2. Model rollback



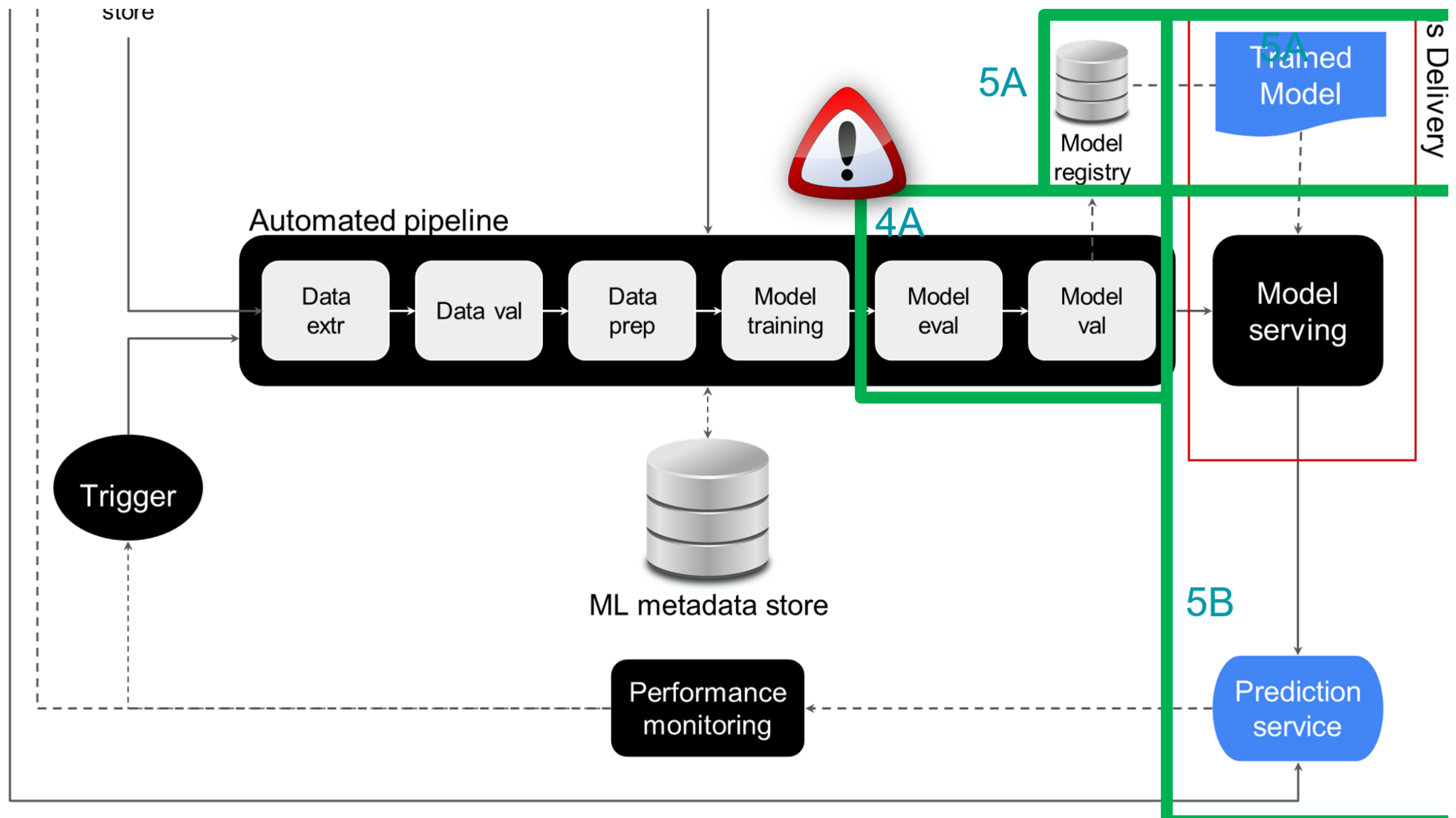
2. Model rollback - Validation fail



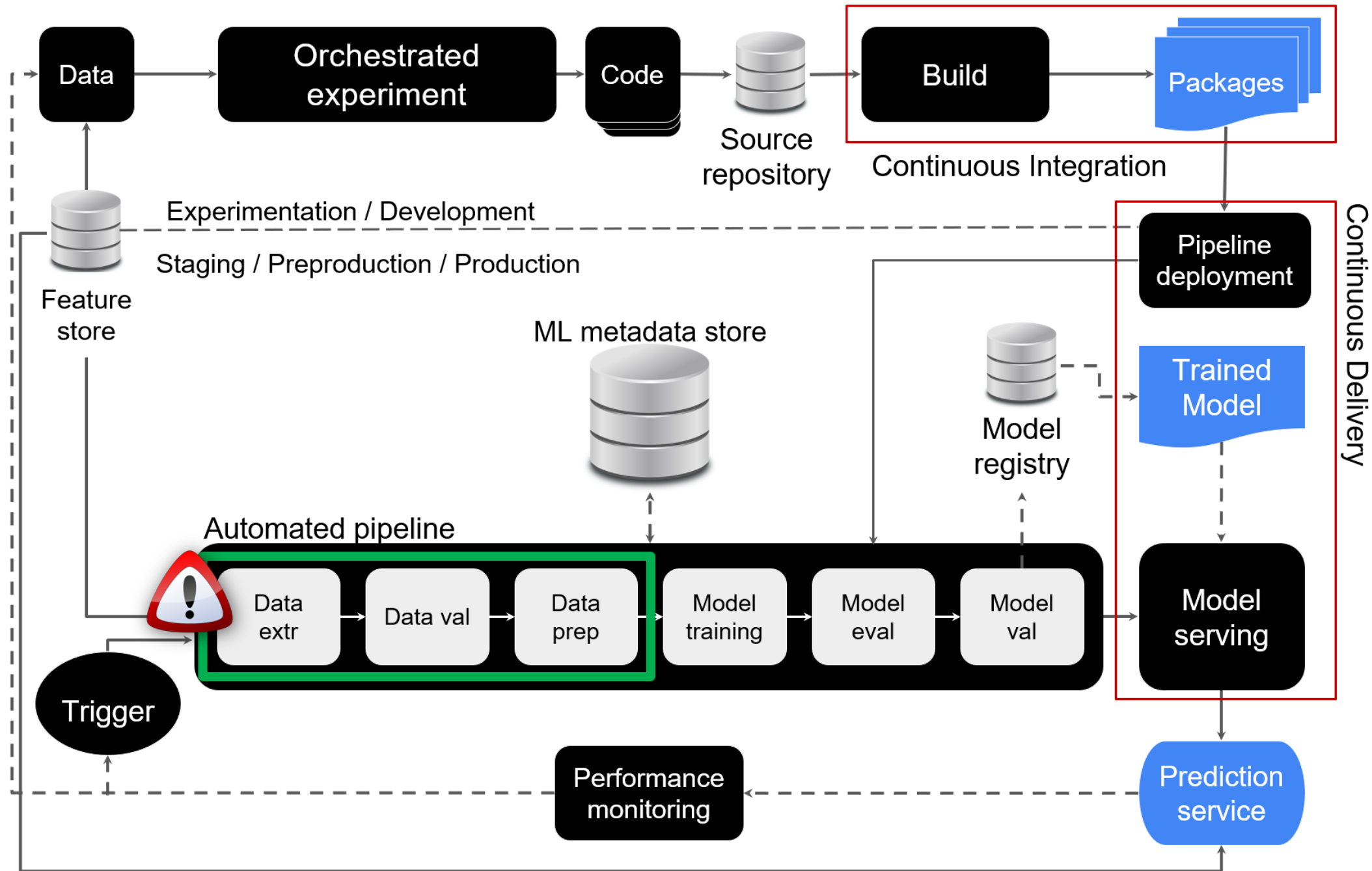
2. Model rollback - Last functional model



2. Model rollback - Redeployment



3. Feature imputation - Data intensive pipeline



3. Feature imputation - Data quality

Feature A	Feature B	• • •	Feature Y	Feature Z
213	Blue		908	Low
234	NaN		NaN	High
132	NaN		731	Medium
182	NaN		NaN	Very Low
190	Red		NaN	Very High
• • •			• • •	
179	NaN		NaN	Low
139	NaN		NaN	Very High
285	NaN		NaN	Very High
401	NaN		NaN	Medium

- Varying levels of data quality
- Some features might fall below a QA threshold

3. Feature imputation - Defective features

Feature A	Feature B		Feature Y	Feature Z
213				Low
234	NaN		NaN	High
132	NaN		731	Medium
182	NaN		NaN	Very Low
190	Red		NaN	Very High
⋮			⋮	
179	NaN		NaN	Low
139	NaN		NaN	Very High
285	NaN		NaN	Very High
401	NaN		NaN	Medium

- Detect failing features
- Apply feature imputation

3. Feature imputation - Potential fixes

- Numerical Values
 - Mean/Median Imputation
 - KNN Imputation
- Categorical Values
- Frequent Category Imputation
- Adding a "Missing" Category

Let's practice!
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Automated testing in MLOps

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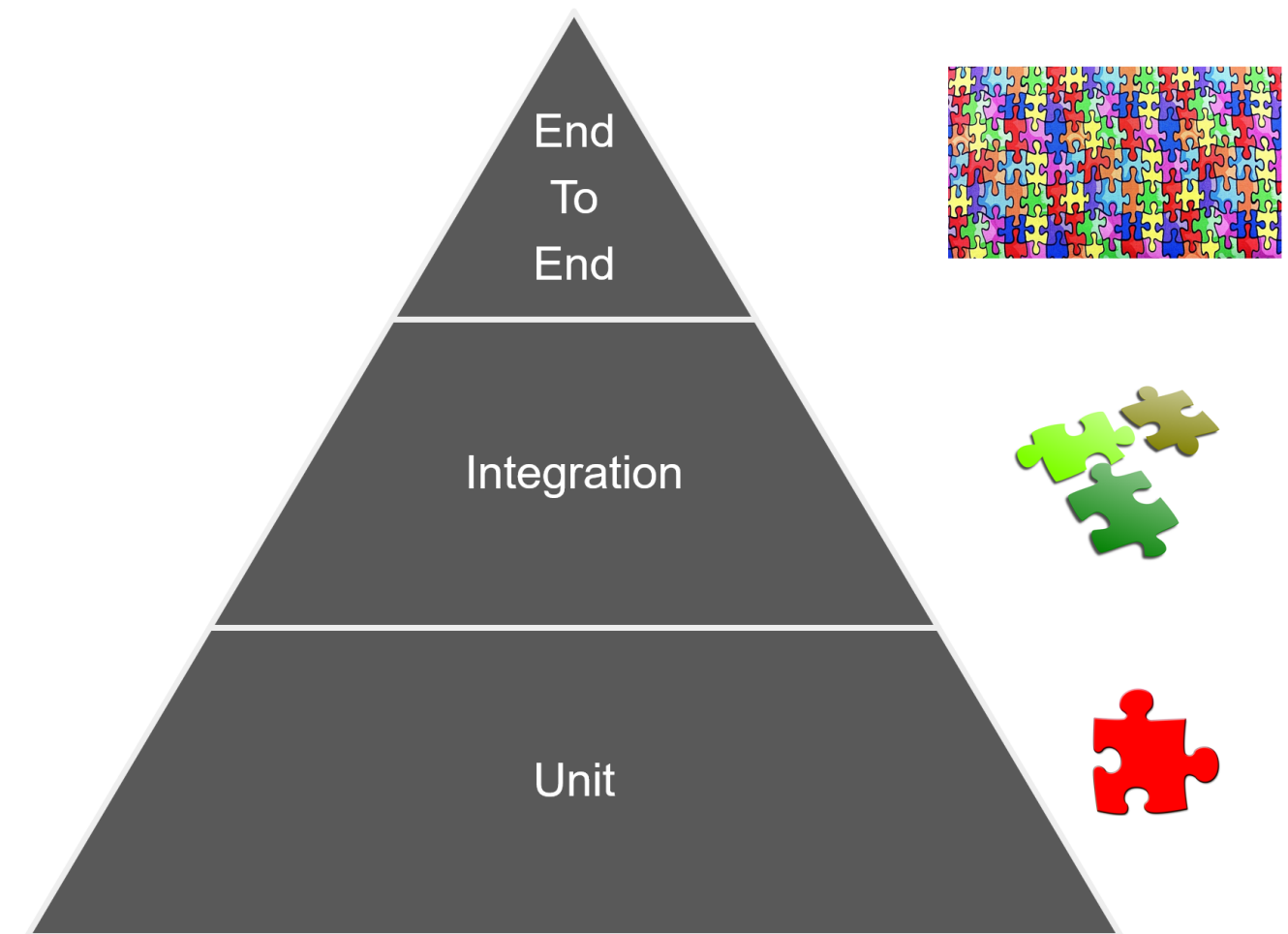
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Senior Consultant - Machine Learning

What is software testing?

The process of evaluating and verifying that a software product or application works as expected

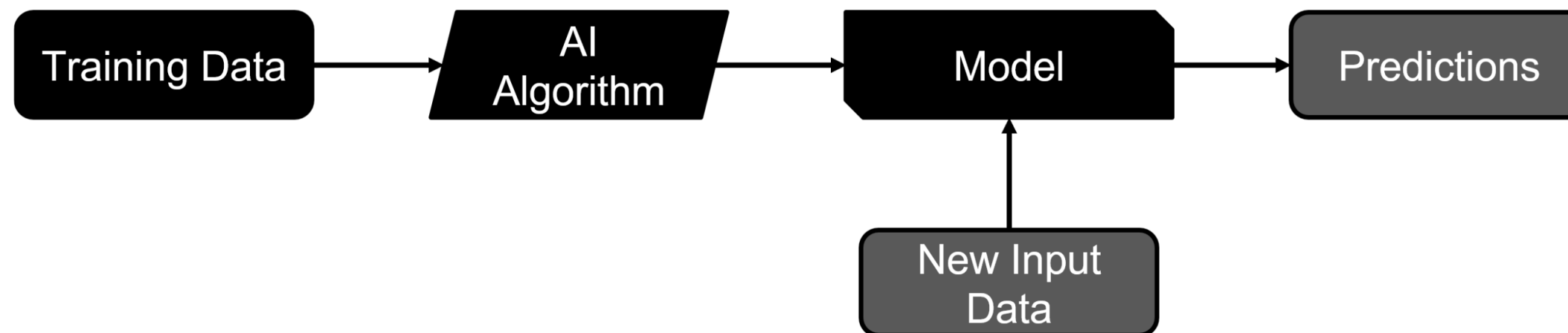
Three most common types of software testing:

- Unit tests
- Integration tests
- End-to-end tests



ML software has a different nature

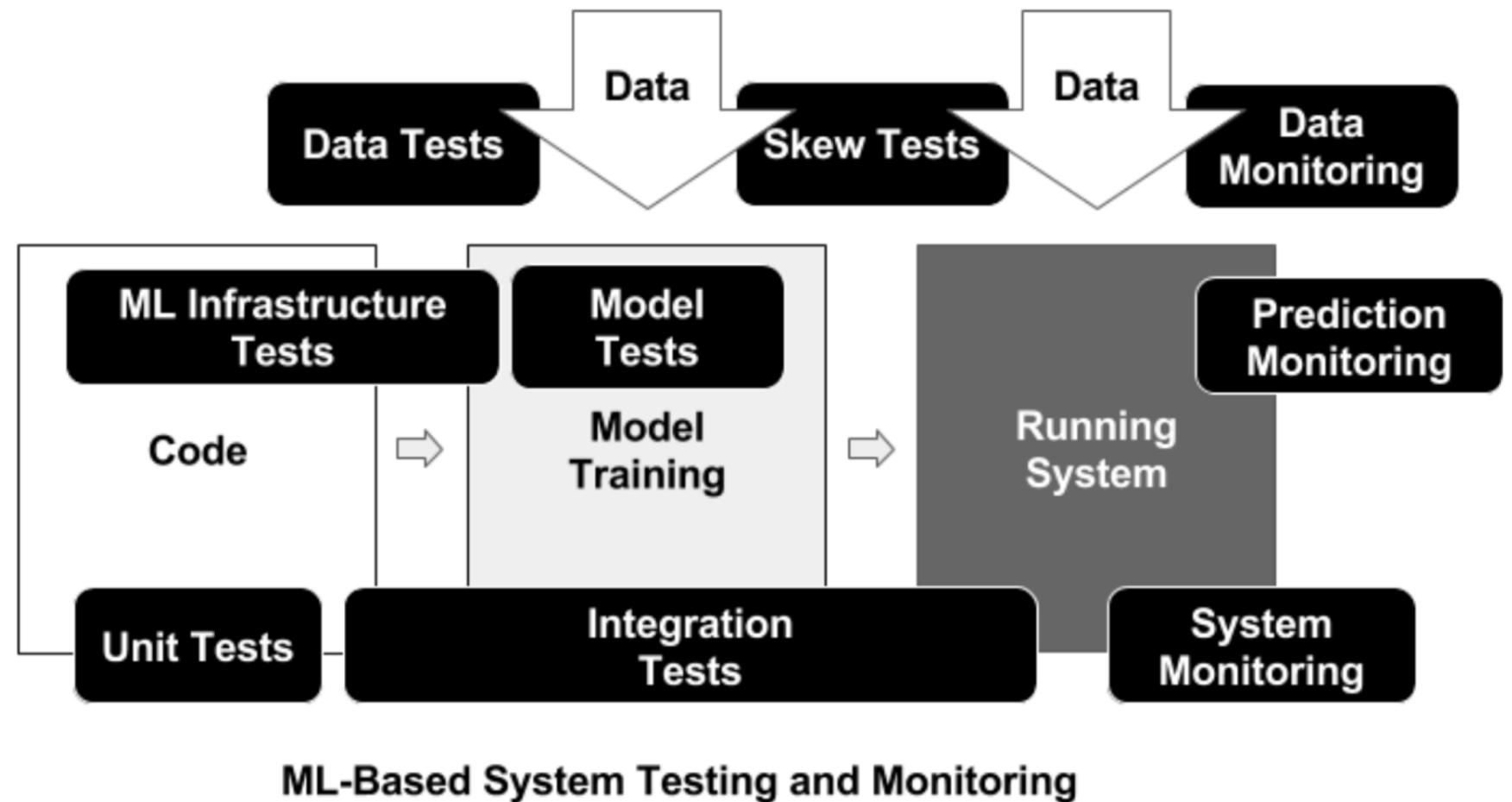
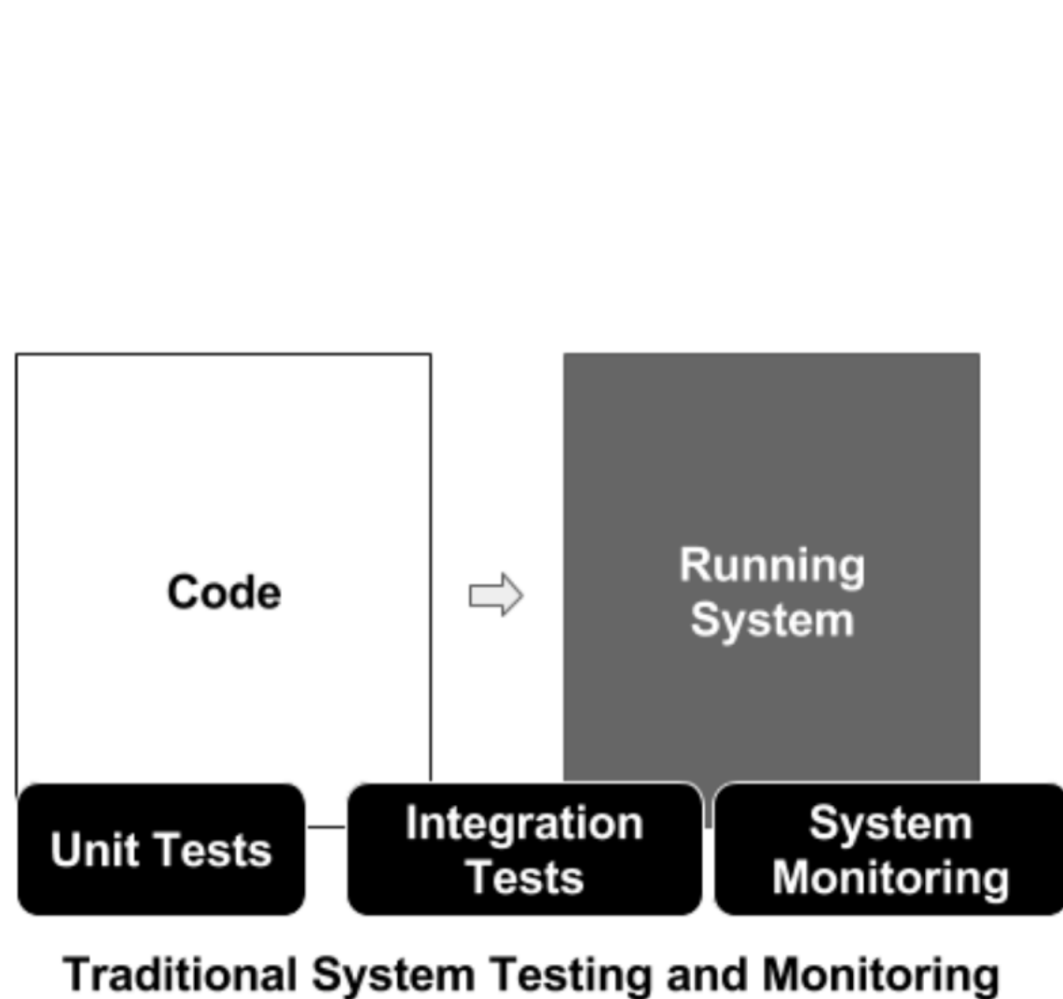
ML APPLICATIONS ARE NOT EXPLICITLY PROGRAMMED



They depend on:

- Data
- Models

Testing in an MLOps system is different



¹ <https://research.google/pubs/pub46555/>

Testing ML systems

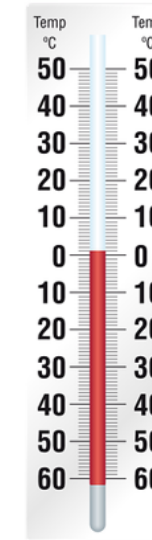
In addition to traditional software testing:

- Data tests
- Model tests
- Pipeline tests

Testing the data

Tests for features and data include:

- Feature expectations
- Value from data feature justify its costs
- Privacy control
- Avoid used of unlawful data



Testing the models

Tests for models include:

- Business & ML metrics correlate
- All hyperparameters have been tuned
- Model overfitting
- Model staleness
- Baseline comparison

Testing ML pipelines

Tests for ML pipelines include:

- Training is reproducible
- Integration testing of the ML pipeline
- Model debuggability

Let's practice!
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Automated hyperparameter tuning

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What is a hyperparameter?

Hyperparameters are tunable values that control the learning process

- Not learned during the training process
- Set before training an ML model

Examples:

- Model architecture in a Neural Network
- Number of branches in a decision tree
- Learning rate

What is hyperparameter tuning?



Hyperparameters



Layers = 5
Neurons = 512
Learning rate = 0.1



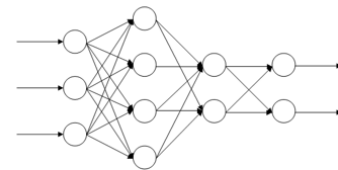
Layers = 5
Neurons = 1024
Learning rate = 0.01



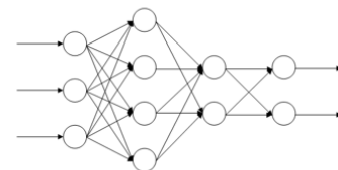
Layers = 6
Neurons = 2048
Learning rate = 0.01



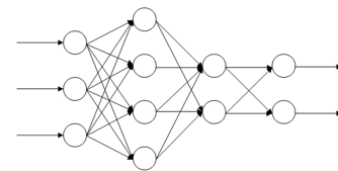
Model Parameters



Weights
optimization



Weights
optimization



Weights
optimization



Scoring

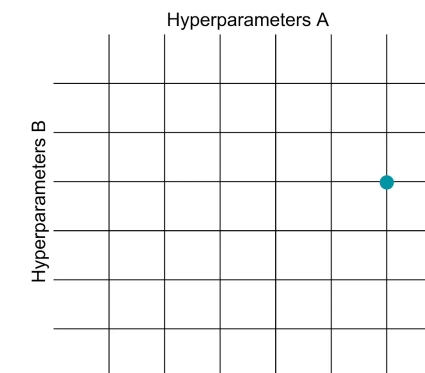
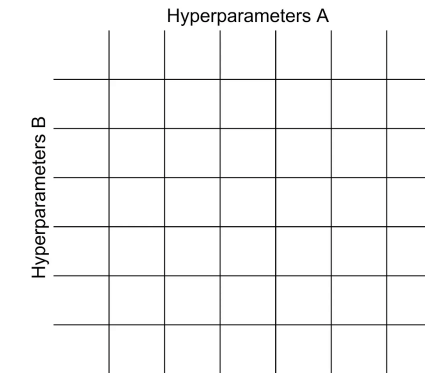
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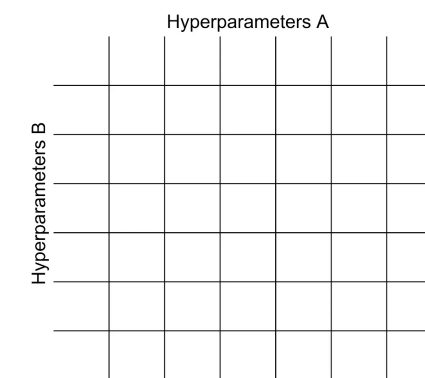
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Hyperparameter tuning methods

- Grid Search
- Random Search
- Bayesian Optimization

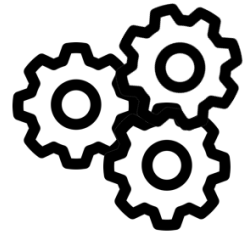


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Automate hyperparameter tuning

AUTOMATE



Hyperparameters



Layers = 5
Neurons = 512
Learning rate = 0.1



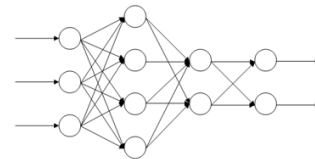
Layers = 5
Neurons = 1024
Learning rate = 0.01



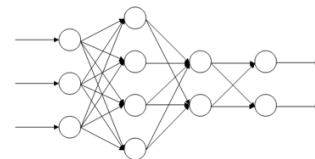
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Neurons = 2048
Learning rate = 0.01



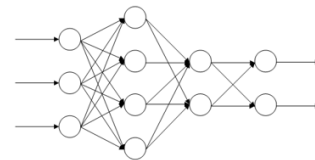
Model Parameters



Weights
optimization



Weights
optimization



Weights
optimization



Scoring

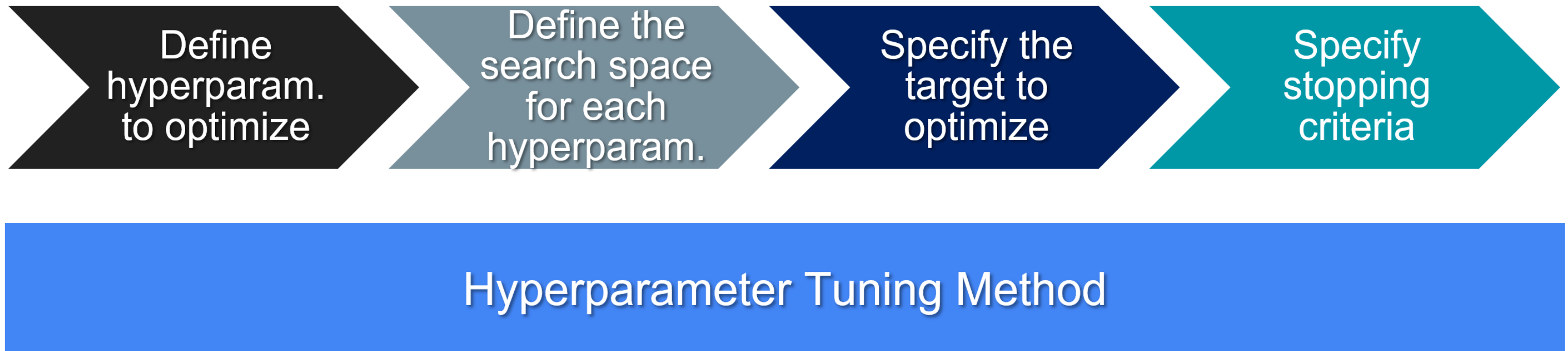
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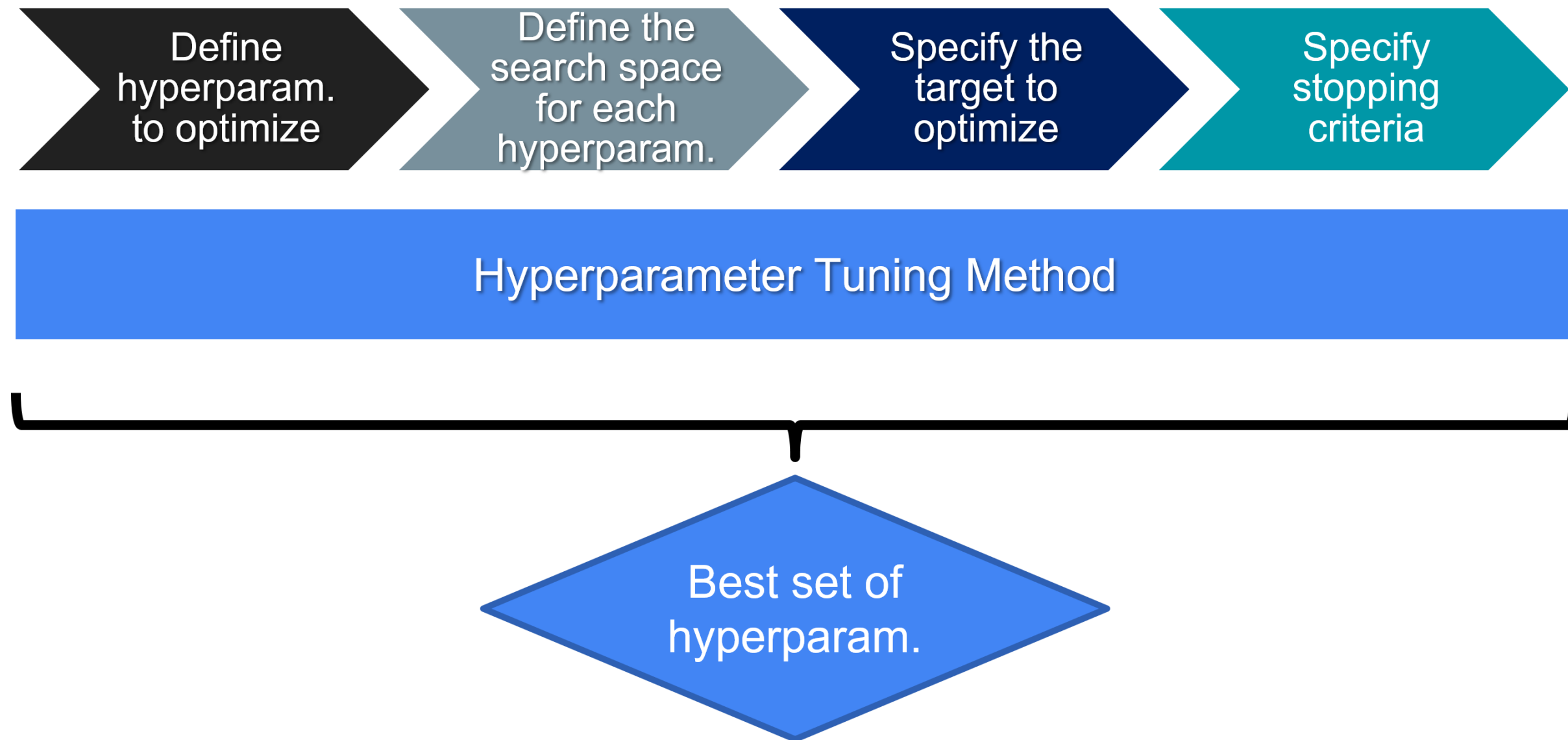
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Automated hyperparameter tuning steps

- Need to define:
 - Set of hyperparameters to optimize
 - Search space for each parameter
 - A performance metrics to optimize
 - Stopping Criteria



Automatically finding the best set of hyperparameters



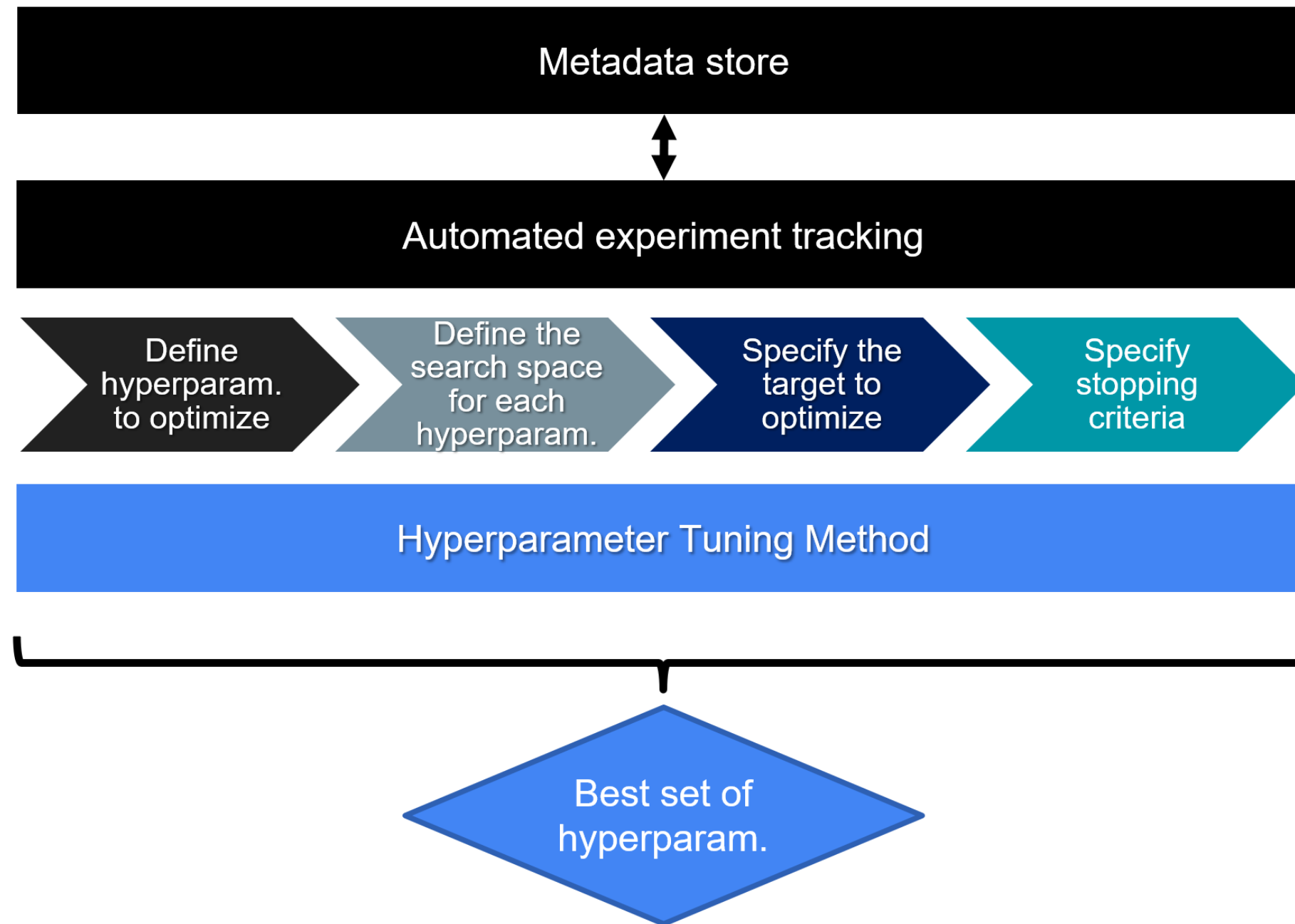
Hyperparameters and environment symmetry

Development &
Experimentation

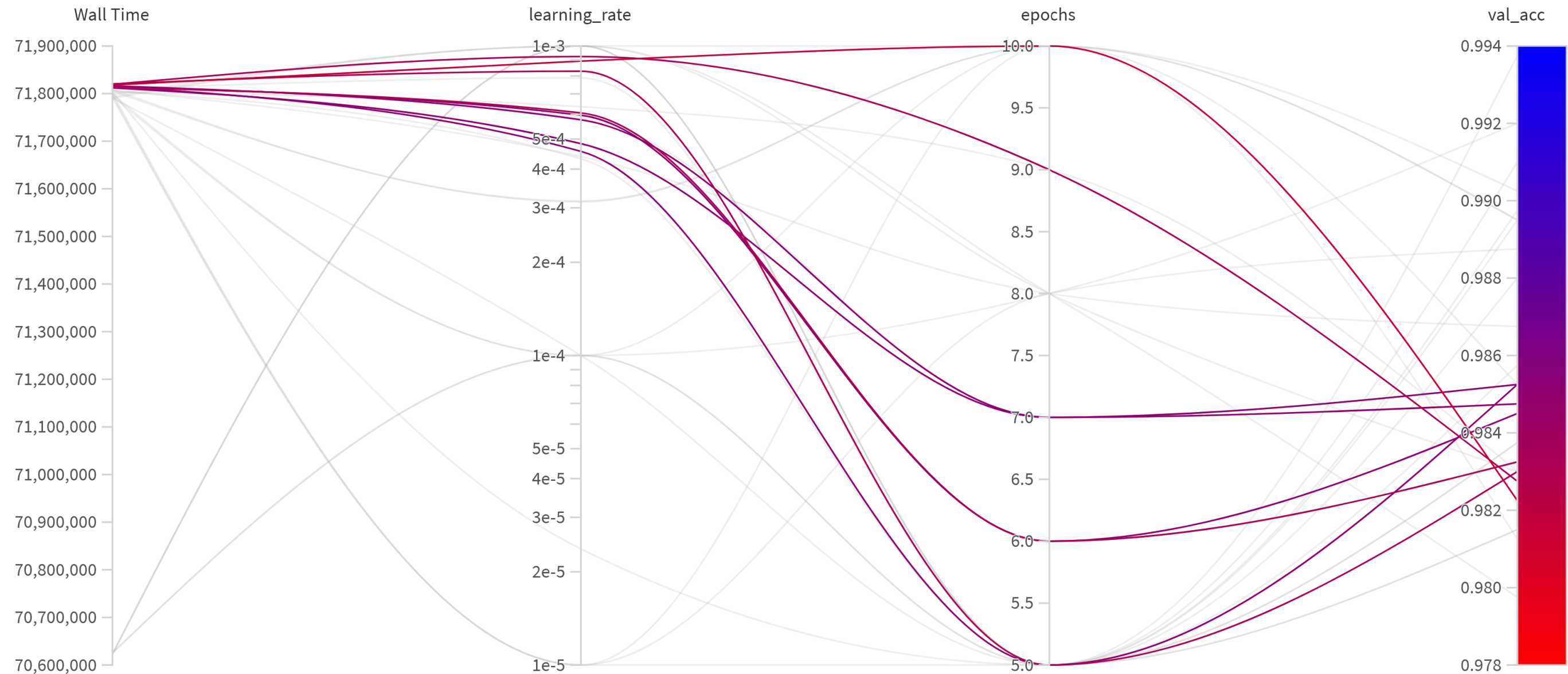
Production

Hyperparameter Tuning

Hyperparameter tuning - Experiment tracking



Example - Hyperparameter visualization



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